



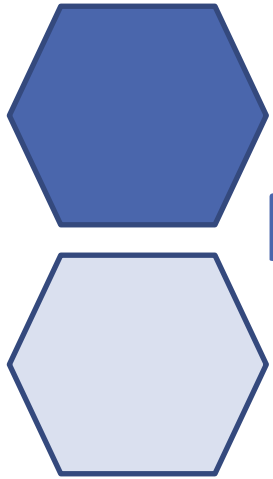
Bioinformatics

Ing. Giulia Fiscon, PhD

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Department of Computer, Control, and Management Engineering
Sapienza University of Rome

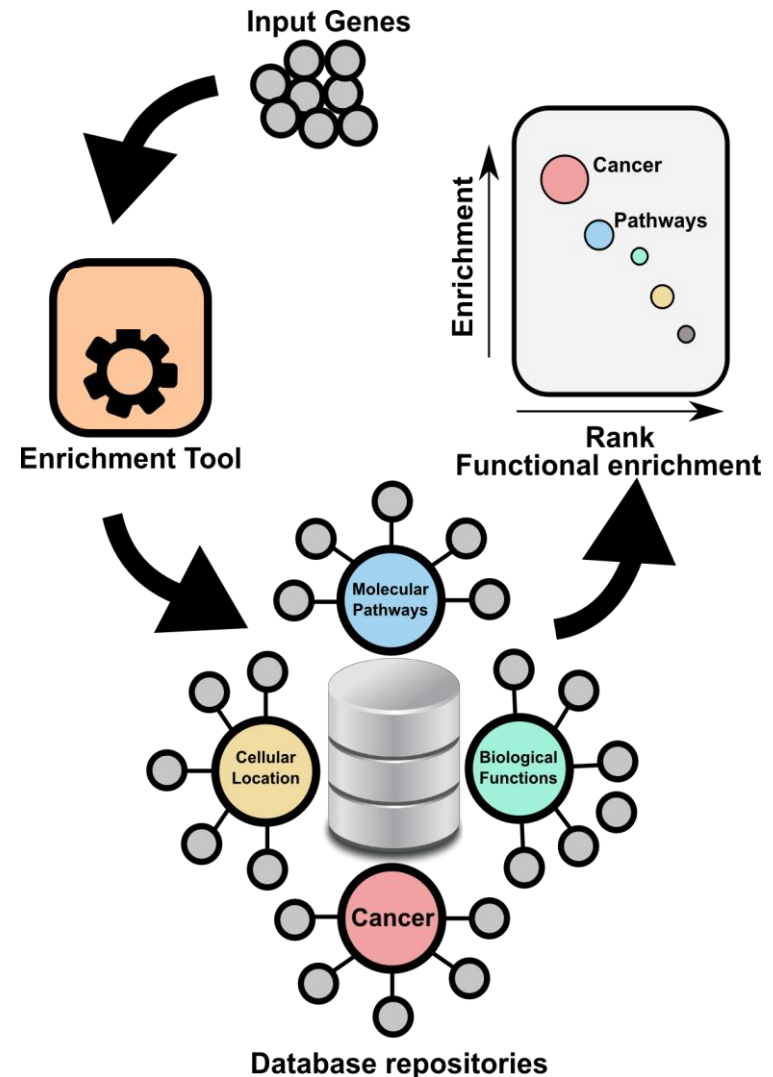
⬡ Functional enrichment analysis and enrichR



Functional enrichment

Functional enrichments

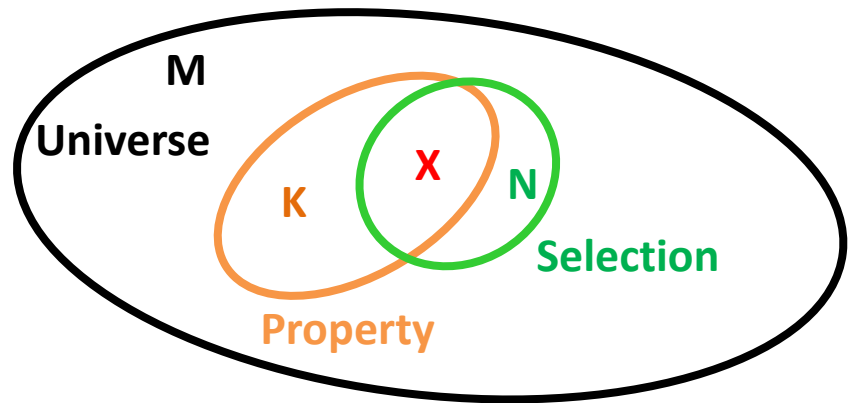
- The existing technologies allow to simultaneously highlight the activity of thousands of genes. Results of these analyses are candidate genes that may be functionally relevant in the examined experimental condition
- The interpretation of such results requires the annotation of the selected genes to functional and structural categories, using appropriate controlled vocabularies, that univocally define their features enabling a standardized analysis
- Using these annotations, we can group the genes of a given list according to their functional and structural features, and study the statistical significance of these categories



Hypergeometric test

Test of over-representation: *which are the significant categories in the list of selected genes?*

- **M**: universe (human genome)
- **K**: set of elements with a given property (all genes annotated for a given functional category)
- **N**: set of elements for which you want to measure the statistical significance of the enrichment in the property K (the list of selected genes)



- | | |
|--|--|
| ■ H₀ : if a gene is annotated for a particular category and belongs to the list of selected genes is due to chance | ■ p-value $\rightarrow 0$ The null hypothesis is rejected and we can conclude that the list of genes is enriched for the considered functional category respect to what one would expect by chance |
| ■ H₁ : the class of selected genes in the tested list is enriched in the considered category. This means that the category is over-represented on the list of selected genes | ■ p-value $\rightarrow 1$ the null hypothesis is accepted, that is list does not enriched for the considered functional category |

Hypergeometric test

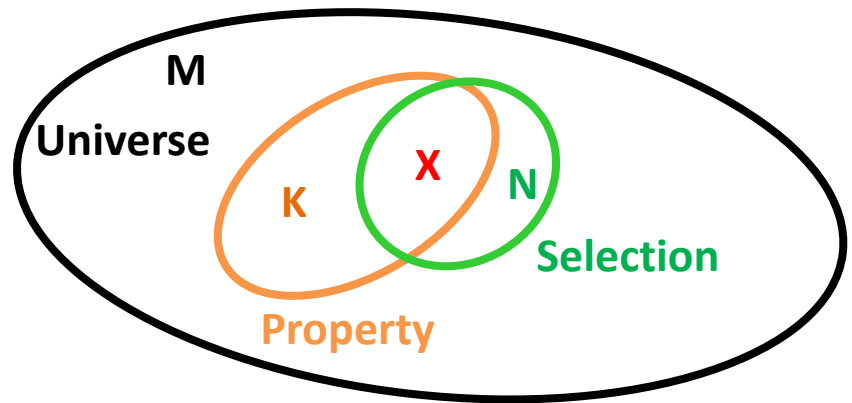
Probability of observing **X elements** with the K property is given by:

$$P(X) = \frac{\binom{K}{X} \binom{M-K}{N-X}}{\binom{M}{N}}$$

$\binom{K}{X}$ number of possible draws of X objects from K objects

$\binom{M-K}{N-X}$ number of possible draws of the remaining $N-X$ objects among the $M-K$ objects

$\binom{M}{N}$ number of possible draws of N objects from M objects



The hypergeometric distribution is a **discrete probability distribution** that calculates the likelihood an event happens X times in N trials when you are sampling from a small population **without replacement**, so describes the probability of successes in draws without replacement

Hypergeometric test

Probability of observing **X elements** with the K property is given by:

$$P(X) = \frac{\overset{\text{\# of successes}}{\binom{K}{X}} \overset{\text{\# of unsuccesses}}{\binom{M-K}{N-X}}}{\underset{\text{Total draws}}{\binom{M}{N}}}$$

Given an urn with two colors of marbles, (M-K) blue and (K) orange. The hypergeo test computes the probability of extracting X number of orange marbles by drawing without replacement N marbles from the urn

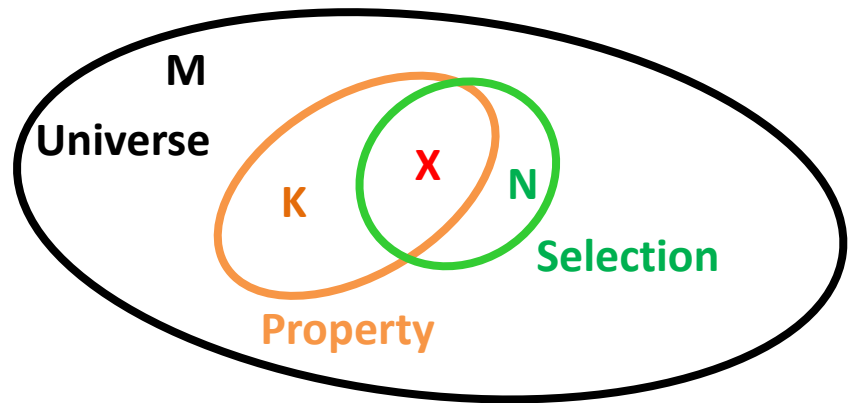
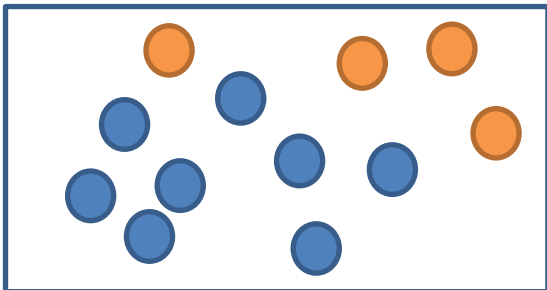
Example:

M = # of total marbles (12)

K = # of orange marbles (4)

M-K = # of blue marbles (8)

N = # of drawn marbles



Example:

M = transcriptome (all human genes)

N = # of selected genes (e.g., DEG)

K = # of genes annotated for a functional category you are testing (e.g., cell cycle)

X = # of genes of your list annotated for cell cycle

The ratio X/N is greater than K/M?

The list of N genes is over-represented in that category

Hypergeometric test

How many beans are in the box?

$$M = 10$$

How many beans are chosen, at random, without replacement?

$$N = 5$$

How many yucky flavors are in the box?

$$K = 4$$

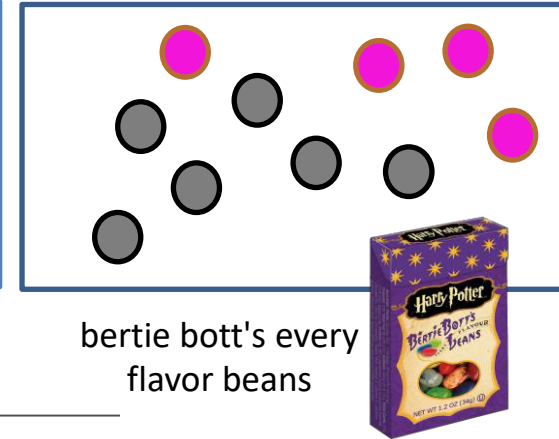
How many yucky flavors to be chosen?

$$X = 2$$

$$P(X) = \frac{\binom{K}{X} \binom{M-K}{N-X}}{\binom{M}{N}}$$

Yucky Tasty

Total

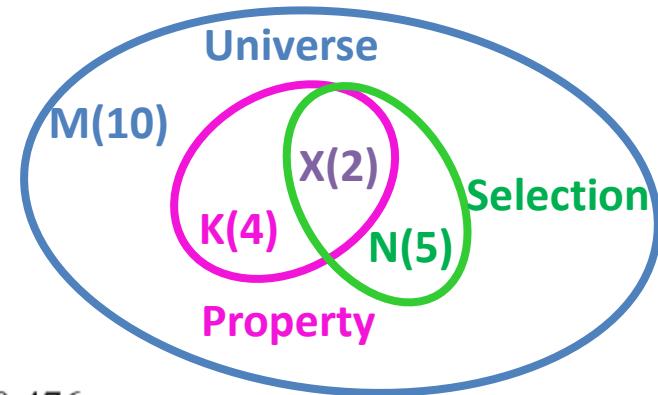


bertie bott's every flavor beans

Probability of extracting 2 (out of 4) yucky beans by drawing without replacement 5 beans

$$P(X) = \frac{\binom{4}{2} \binom{10-4}{5-2}}{\binom{10}{5}}$$

$$P(X) = \frac{\binom{4}{2} \binom{6}{3}}{\binom{10}{5}} = \frac{\left(\frac{4!}{2!(4-2)!} \right) \left(\frac{6!}{3!(6-3)!} \right)}{\left(\frac{10!}{5!(10-5)!} \right)} = \frac{(6)(20)}{252} = 0.476$$

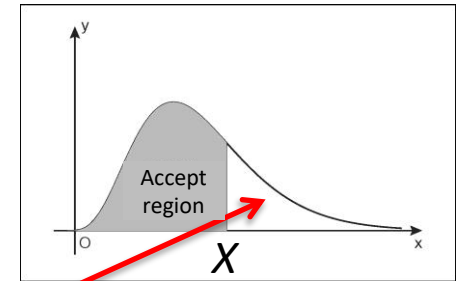


Hypergeometric test

Test of over-representation: *which are the significant categories in the list of selected genes?*

Probability of observing **X or more than elements** with the K property is given by:

$$\Pr = 1 - \sum_{i=0}^{X-1} \frac{\binom{K}{i} \binom{M-K}{N-i}}{\binom{M}{N}} = \sum_{i=X}^N \frac{\binom{K}{i} \binom{M-K}{N-i}}{\binom{M}{N}}$$

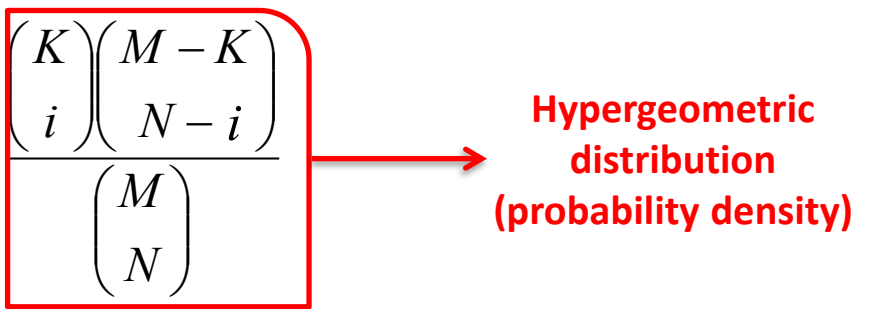


- **H₀**: if a gene is annotated for a particular category and belongs to the list of selected genes is due to chance
- **H₁**: the class of selected genes in the tested list is enriched in the considered category. This means that the category is over-represented on the list of selected genes
- p-value $\rightarrow 0$ The null hypothesis is rejected and we can conclude that the list of genes is enriched for the considered functional category respect to what one would expect by chance
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Hypergeometric test

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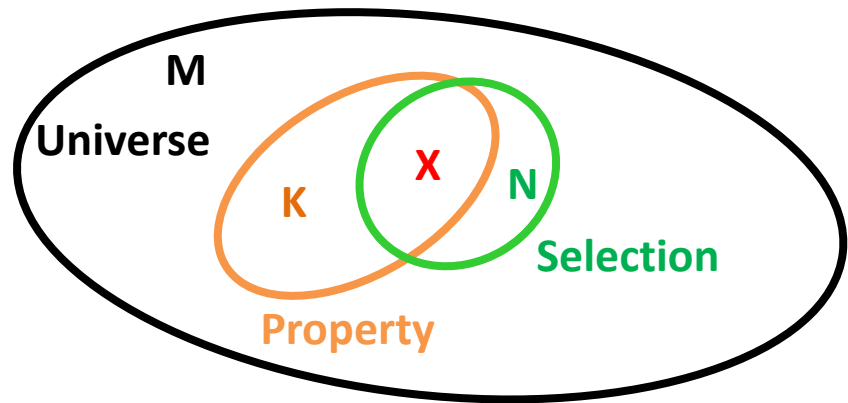
Hypergeometric distribution (probability density)

- **H₀**: if a gene is annotated for a particular category and belongs to the list of selected genes is due to chance
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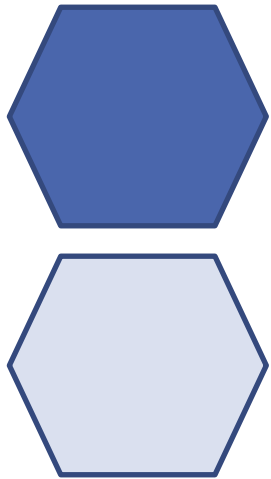
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enrichr

interactive and collaborative gene list
enrichment analysis tool

<https://amp.pharm.mssm.edu/Enrichr/>



Input data

Choose an input file to upload. Either in BED format or a list of genes.

Try an example [BED file](#).

Nessun file selezionato

Paste a list of valid Entrez gene symbols on each row in the text-box below. [Try a gene set example](#).

0 gene(s) entered

Enter a brief description for the list in case you want to share it. (Optional)

☐ **Contribute**

Please acknowledge Enrichr in your publications by citing the following references:

Chen EY, Tan CM, Kou Y, Duan Q, Wang Z, Meirelles GV, Clark NR, Ma'ayan A. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC Bioinformatics*. 2013;128(14).

Kuleshov MV, Jones MR, Rouillard AD, Fernandez NF, Duan Q, Wang Z, Koplev S, Jenkins SL, Jagodnik KM, Lachmann A, McDermott MG, Monteiro CD, Gundersen GW, Ma'ayan A. Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. *Nucleic Acids Research*. 2016; gkw377.



Enrichr

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21,483,193 lists analyzed

318,171 terms

156 libraries

Analyze[What's new?](#)[Libraries](#)[Gene search](#)[About](#)[Help](#)

Upload the
input list of
genes

Input data

Choose an input file to upload. Either in BED format or a list of genes.

Try an example [BED file](#).

Nessun file selezionato

Or
Paste the list of
genes

Paste a list of valid Entrez gene symbols on each row in the text-box below. [Try a gene set example](#).

```
ADORA2B
AGPAT9
AIM1
AKR1C2
ALPK2
AMIGO2
ANLN
ANXA2P3
AOX1
APOBEC3B
```

171 gene(s) entered

Enter a brief description for the list in case you want to share it. (Optional)

☐ Contribute

Please acknowledge Enrichr in your publications by citing the following references:

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Give a name to
your input list
(optional)

modEnrichr

A suite of gene list enrichment analysis tools

FlyEnrichr

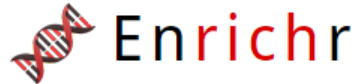
YeastEnrichr

WormEnrichr

FishEnrichr

[Click here to raise an issue on GitHub](#)

Browsing the results



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[Cell Types](#)

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[Legacy](#)

[Crowd](#)

Description Sample gene list (262 genes)



ChEA 2016



FOXO1_25889361_ChIP-Seq_OE33_AND_U2OS
FOXO1_23109430_ChIP-Seq_U2OS_Human
E2F4_17652178_ChIP-Seq_URKAT_Human
AR_21909140_ChIP-Seq_LNCAP_Human
KDM5B_21448134_ChIP-Seq_MESCs_M

ENCODE and ChEA Consensus TFs from



E2F4_ENCODE
FOXO1_ENCODE
SIN3A_ENCODE

ARCHS4 TFs Coexp



HMGB2_human_tf_ARCHS4_coexpression
E2F8_human_tf_ARCHS4_coexpression
E2H2_human_tf_ARCHS4_coexpression
human_tf_ARCHS4_coexpression
human_tf_ARCHS4_coexpression

TF Perturbations Followed by Expression

ZNF750_KD_HUMAN_GSE38039_CREEDSID_GEN
ZNF750_KD_HUMAN_GSE38039_CREEDSID_GEN
FLI1_KD_HUMAN_GSE27524_CREEDSID_GEN
FLI1_KD_HUMAN_GSE27524_CREEDSID_GEN
FLI1_KD_HUMAN_GSE27524_CREEDSID_GEN

YBX1_human
YBX1_human
E2F4_human
E2F3_human

TF Submissions TF- Cooccurrence



DEPDC1
DEPDC1
FOXO1
E2F8

TRANSFAC and JASPAR PWMs



E2F4 (human)
NFYA (human)
E2F1 (human)
NFIC (human)
RUNX1 (human)

Epigenomics Roadmap HM ChIP-seq



H3K4me1 Skeletal Muscle
H3K4me1 Colon Smooth Muscle
H3K4me3 Mesenchymal Stem Cell Derived A
H3K4me1 Brain Inferior Temporal Lobe
H3K4me1 Stomach Smooth Muscle

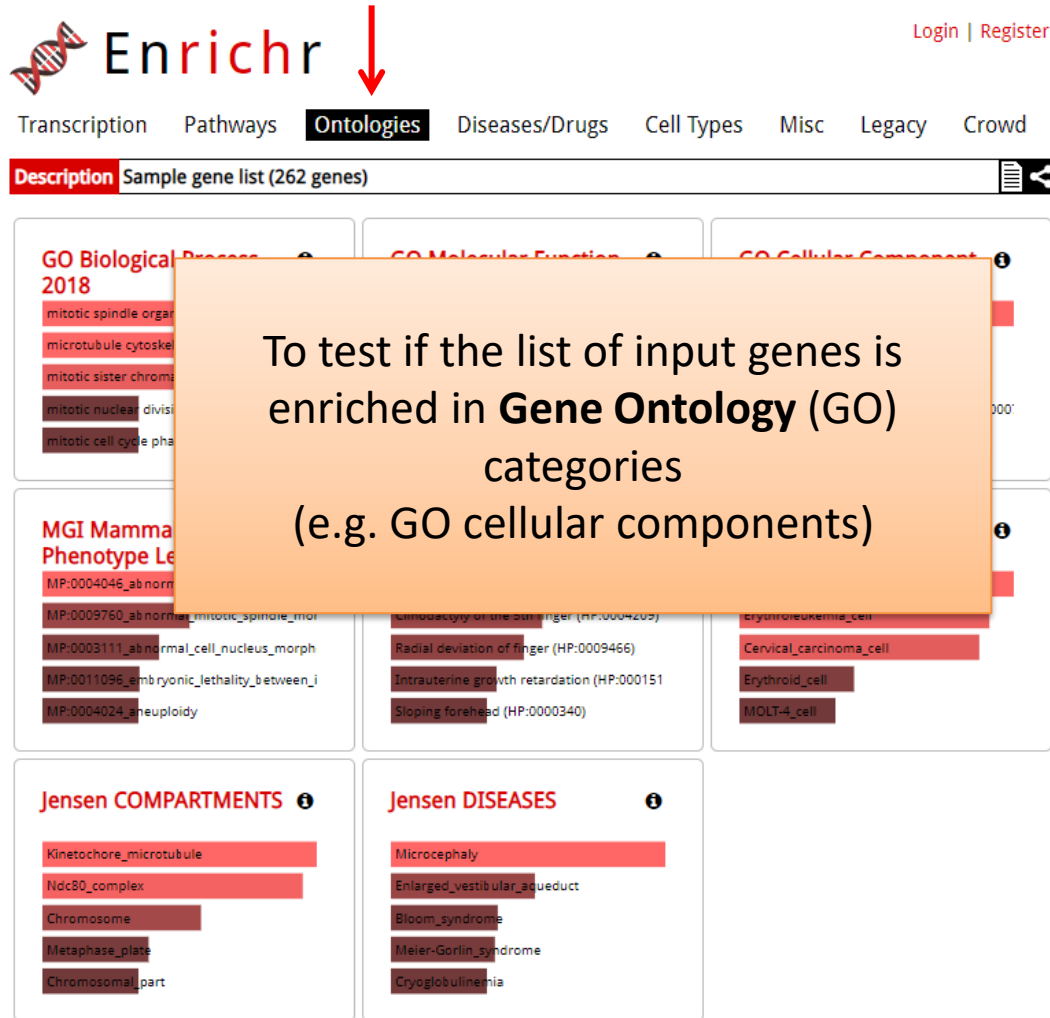
TargetScan microRNA 2017



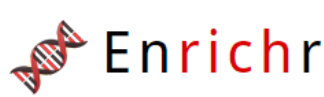
hsa-miR-1204
hsa-miR-3677-3p
hsa-miR-3648
mmu-miR-471-5p
mmu-miR-5126

On the results page, the analysis is divided into different categories of enrichment

Gene Ontology (GO) results



Gene Ontology (GO)



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Description Sample gene list (262 genes)



GO Biological Process 2018



mitotic spindle organization (GO:0007052)
microtubule cytoskeleton organization invol
mitotic sister chromatid segregation (GO:00
mitotic nuclear division (GO:0140014)
mitotic cell cycle phase transition (GO:0044;

GO Molecular Function 2018



microtubule binding (GO:0008017)
microtubule motor activity (GO:0003777)
tubulin binding (GO:0015631)
motor activity (GO:0003774)
3'-5' DNA helicase activity (GO:0043138)

GO Cellular Component 2018



spindle (GO:0005819)
mitotic spindle (GO:0072686)
spindle microtubule (GO:0005876)
chromosome, centromeric region (GO:0000
spindle midzone (GO:0051233)



MGI Mammalian Phenotype Level 4 2019



MP:0004046_abnormal_mitosis
MP:0009760_abnormal_mitotic_spindle_mor
MP:0003111_abnormal_cell_nucleus_morph
MP:0011096_embryonic_lethality_between_i
MP:0004024_anuploidy

Human Phenotype Ontology



Cafe-au-lait spot (HP:0000957)
Clinodactyly of the 5th finger (HP:0004209)
Radial deviation of finger (HP:0009466)
Intrauterine growth retardation (HP:000151
Sloping forehead (HP:0000340)

Jensen TISSUES



Vegetative_cell
Erythroleukemia_cell
Cervical_carcinoma_cell
Erythroid_cell
MOLT-4_cell

Jensen COMPARTMENTS



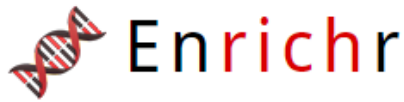
Kinetochore_microtubule
Ndc80_complex
Chromosome
Metaphase_plate
Chromosomal_part

Jensen DISEASES



Microcephaly
Enlarged_vestibular_aqueduct
Bloom_syndrome
Meier-Gorlin_syndrome
Cryoglobulinemia

Understanding the results



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Description No description available (171 genes)

GO Biological Process 2018

GO Molecular Function 2018

GO Cellular Component 2018

Bar Graph

Table

Clustergram



Click the bars to sort. Now so

p-value: 4.545e-7
q-value: 0.0002027
odds ratio: 4.91
combined score: 71.77

SVG PNG JPG

focal adhesion (GO:0005925)

endoplasmic reticulum lumen (GO:0005788)

integral component of plasma membrane (GO:0005887)

actin cytoskeleton (GO:0015629)

Pointing each row, the enrichment scores of each term are displayed

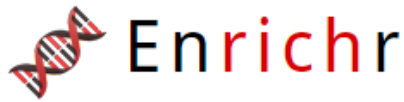
You can sort the graph by the term, p-value, z-score, or combined score presented in the data table.

You can export the bar graph as a figure by clicking on one of the image format buttons to the top right of the bar graph.

Bar graph:

The length of the bar represents the significance of that specific gene-set or term. In addition, the brighter the color, the more significant that term is.

Understanding the results



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Transcription Pathways **Ontologies** Diseases/Drugs Cell Types Misc Legacy Crowd

Description No description available (171 genes)

GO Biological Process 2018

GO Molecular Function 2018

GO Cellular Component 2018

Bar Graph **Table** Clustergram

Hover each row to see the overlapping genes.

10 entries per page

Enrichment scores

Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	focal adhesion (GO:0005925)	4.545e-7	0.0002027	4.91	71.77
2	platelet alpha granule membrane (GO:0031092)	0.009085	0.5788	13.76	64.69
3	endoplasmic reticulum lumen (GO:0005788)	0.000003915	0.0008730	5.18	64.48
4	FACIT collagen trimer (GO:0005593)	0.05022	1.000	19.49	58.31
5	protein kinase complex (GO:1902911)	0.05022	1.000	19.49	58.31
6	multivesicular body membrane (GO:0032585)	0.05834	1.000	16.71	47.48
7	microtubule minus-end (GO:0036449)	0.06640	1.000	14.62	39.65
8	myosin filament (GO:0032982)	0.08230	1.000	11.70	29.21
9	caveola (GO:0005901)	0.01282	0.7147	6.16	26.82
10	cell cortex region (GO:0099738)	0.09015	1.000	10.63	25.58

Enriched terms

Showing 1 to 10 of 114 entries | [Export entries to table](#)

Terms marked with an * have an overlap of less than 5

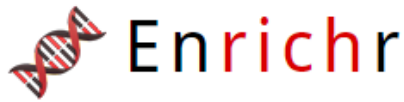
[Previous](#) [Next](#)

- **p-value (p)**: computed by using the Fisher's exact test (hypergeometric test)
- **adjusted p-value**: computed by using the Benjamini-Hochberg method (FDR) for correction for multiple hypotheses testing.
- **Odds ratio (z)**: zscore computed with respect to precomputed distribution for many random input gene lists for each term in the gene set library.
- **Combined score**:

$$c = |\ln(p) * z|$$

$$\frac{k/u}{x/n}$$

Understanding the results



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GO Biological Process 2018

GO Molecular Function 2018

GO Cellular Component 2018

[Bar Graph](#) **[Table](#)** [Clustergram](#) [Settings](#) [Info](#)

Hover each row to see the overlapping genes.

10 entries per page

Search:

Index	Name	Adjusted p-value	Odds Ratio	Combined score	
1	focal adhesion (GO:0005925) SVIL, MME, ITGA3, CAV1, FHL2, PLAUR, RRAS2, NEXN, CNN2, PROCR, DAB2, MYH9, FLNB, ITGA5, EPHA2	4.545e-7	0.0002027	4.91	71.77
2	platelet alpha granule membrane (GO:0031092)	0.009085	0.5788	13.76	64.69
3	endoplasmic reticulum lumen (GO:0005788)	0.000003915	0.0008730	5.18	64.48
4	FACT complex (GO:0030698)	1.000	19.49	58.31	
5	prolactin receptor activity (GO:0004866)	1.000	19.49	58.31	
6	mu-opioid receptor activity (GO:0004866)	1.000	16.71	47.48	
7	microtubule minus-end (GO:0036449)	0.06640	1.000	14.62	39.65
8	myosin filament (GO:0032982)	0.08230	1.000	11.70	29.21
9	caveola (GO:0005901)	0.01282	0.7147	6.16	26.82
10	cell cortex region (GO:0099738)	0.09015	1.000	10.63	25.58

Pointing each row, the input genes related to that term are displayed

The list of input genes will be statistically enriched in terms with p-value < 0.05

Showing 1 to 10 of 114 entries | [Export entries to table](#)

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Terms marked with an * have an overlap of less than 5

Understanding the results

GO Cellular Component 2018

Hover each row to see the overlapping genes.

10 ▾ entries per page

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Showing 1 to 10 of 114 entries | [Export entries to table](#)

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Bar Graph

Table

Clustergram

you can filter the results by searching for a specific term

you can sort the table by the term, p-value, z-score, or combined score

You can also get the table information in tab-delimited format by clicking on the "Export entries to table" button.

Exporting TABLE

Functional category

pvalue

Adjusted pvalue

Genes of your input list
annotated in that category

	A	B	D	E	F	G	H	I	J	K	L	M	N
1	Term	Overlap	P-value	Adjusted P-value	Old P-value	Old Adjusted Odds Ratio	Combined Significance	Genes					
2	focal adhesion (GO:0005925)	15/357	4.54E-07	2.03E-04	0	0	4.9142464	71.7684844	SVIL;MME;ITGA3;CAV1;FHL2;PLAUR;RRAS2;NEXN;CNN2;PROCR;DAB2;MYH9;FLN				
3	endoplasmic reticulum lumen (GO:0005788)	12/271	3.91E-06	8.73E-04	0	0	5.178999158	64.4824702	EVA1A;COL5A1;STC2;COL12A1;WNT5A;GPX8;PLAUR;COL6A3;PRSS23;THBS1;CYR				
4	integral component of plasma membrane (GO:0005887)	27/1464	1.22E-04	0.018124865	0	0	2.157031924	19.4395539	PTGFR;TENM3;IRS1;SGMS2;GPR176;SLC16A7;STXBP5;CD55;MICA;IL13RA1;MME;				
5	actin cytoskeleton (GO:0015629)	8/295	0.003951	0.440583917	0	0	3.171771236	17.5515637	ANLN;SVIL;SMTN;MLPH;CDC42EP3;PAWR;MYH9;FLNB				
6	cytoskeleton (GO:0005856)	11/521	0.0053	0.472745721	0	0	2.469385235	12.9397729	CNN2;ANLN;SVIL;SMTN;MVP;PLK1;CDC42EP3;MYH9;FLNB;HMMR;SKA1				
7	platelet alpha granule (GO:0031091)	apr-91	0.007746	0.575752103	0	0	5.141057773	24.9888186	CD109;PHACTR2;VEGFC;THBS1				
8	platelet alpha granule membrane (GO:0031092)	feb-17	0.009085	0.578823918	0	0	13.75988992	64.6875234	CD109;PHACTR2				
9	caveola (GO:0005901)	mar-57	0.01282	0.714725707	0	0	6.155740228	26.8189218	IRS1;CAV1;MYOF				
10	dendrite (GO:0030425)	5/216	0.038295	1	0	0	2.707385748	8.83264913	MLPH;MME;BDNF;PLK2;RAB27A				
11	contractile actin filament bundle (GO:0097517)	feb-41	0.047948	1	0	0	5.705320211	17.3307355	CNN2;MYH9				
12	stress fiber (GO:0001725)	feb-41	0.047948	1	0	0	5.705320211	17.3307355	CNN2;MYH9				
13	FACIT collagen trimer (GO:0005593)	01-giu	0.050222	1	0	0	19.49317739	58.3100639	COL12A1				

GO Cellular Component 2018

Bar Graph

Table

Clustergram



Enriched Terms are the columns, input genes are the rows, and cells in the matrix indicate if a gene is associated with a term.



Row Order

Cluster

Sum

Column Order

Cluster

Sum

Gene

Search

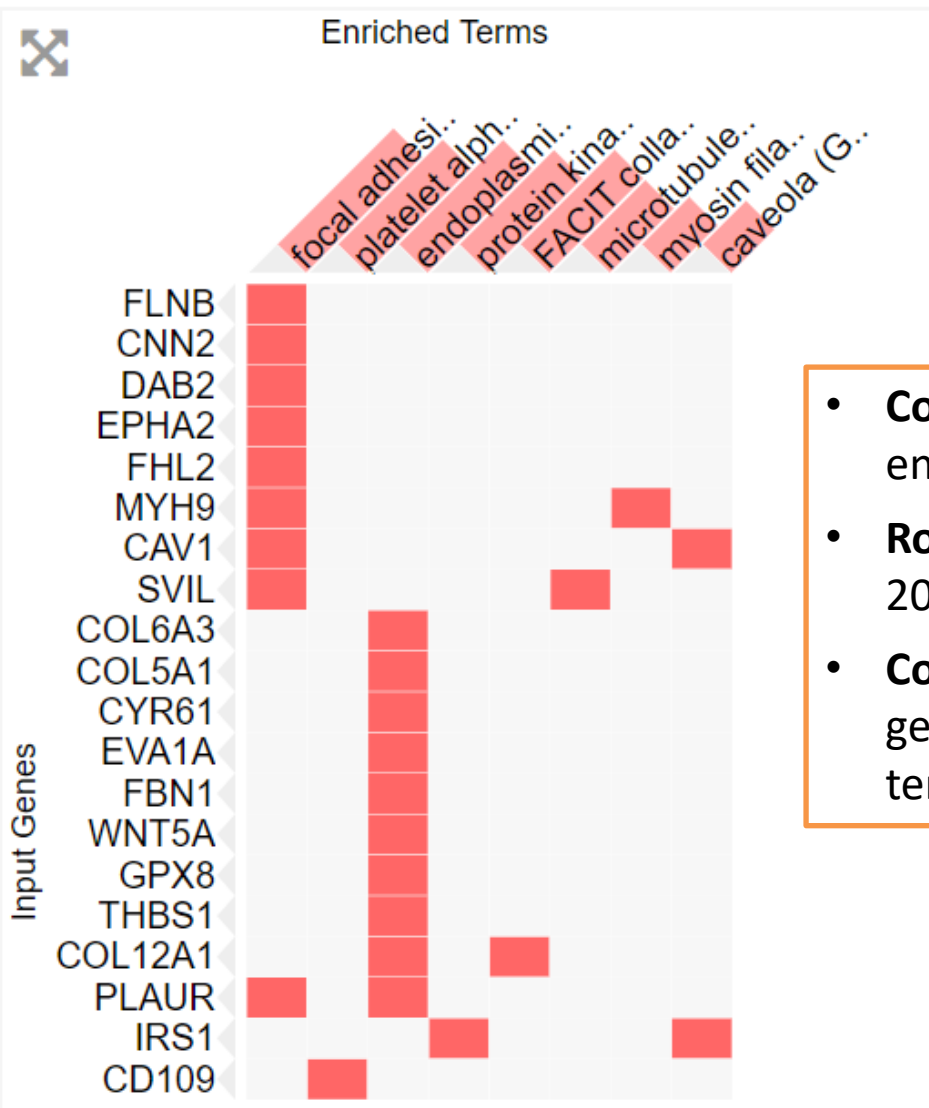
Combined Score

P-Value

Z-score

Top Enriched Terms: 10

Top rows sum: 20 rows



- **Columns: top-10** enriched terms
- **Rows: input genes** (first 20)
- **Colored cells:** indicate if a gene is associated with term

Pathways

 **Enrichr** [Login](#) | [Register](#)

Transcription **Pathways** Ontologies Diseases/Drugs Cell Types Misc Legacy Crowd

Description No description available (171 genes)  

To test if the list of input genes is enriched in **pathways** stored in various databases (e.g. KEGG pathways)

WikiPathways 2019 Human	WikiPathways 2019	KEGG 2019 Human
Canonical and Non-Canonical Wnt Signaling Pathway		
miR-509-3p alteration of YAP1		
Hippo-Merlin Signaling Dysregulation		
PI3K-Akt Signaling Pathway		
miRNA targets in ECM and		

ARCHS4 Kinases Co-expression		
MYLK_human_kinase_ARCHS4_coexpression	Proteoglycans in cancer	TGF beta signaling pathway_Homo sapiens
NEK7_human_kinase_ARCHS4_coexpression	ECM-receptor interaction	Insulin Signaling Pathway_Homo sapiens_h
DDR2_human_kinase_ARCHS4_coexpression	Hematopoietic cell lineage	Multiple antiapoptotic pathways from IGF-1
ABL2_human_kinase_ARCHS4_coexpression	Focal adhesion	The IGF-1 Receptor and Longevity_Homo sapiens
AXL_human_kinase_ARCHS4_coexpression	PI3K-Akt signaling pathway	Bone Remodeling_Homo sapiens_h_ranklPat

Reactome 2016	HumanCyc 2016	NCI-Nature 2016
Elastic fibre formation_Homo sapiens_R-HSA4	allopregnanolone biosynthesis_Homo sapiens	Beta3 integrin cell surface interactions_Homo sapiens
Extracellular matrix organization_Homo sapiens	pyrimidine deoxynucleosides salvage_Homo sapiens	Beta5 beta6 beta7 and beta8 integrin cell surface interactions_Homo sapiens
Crosslinking of collagen fibrils_Homo sapiens	urea cycle_Homo sapiens_PWWY-4984	Validated transcriptional targets of Tap63 in
Collagen formation_Homo sapiens_R-HSA-1+	androgen biosynthesis_Homo sapiens_PWWY	Beta1 integrin cell surface interactions_Homo sapiens
Assembly of collagen fibrils and other multi	nicotine degradation III_Homo sapiens_PWWY	Integrins in angiogenesis_Homo sapiens_2d

Pathways

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[Transcription](#) **[Pathways](#)** [Ontologies](#) [Diseases/Drugs](#) [Cell Types](#) [Misc](#) [Legacy](#) [Crowd](#)

Description No description available (171 genes)  

WikiPathways 2019 Human 

- Canonical and Non-Canonical TGF- β signaling
- miR-509-3p alteration of YAP1/ECM axis WP:
- Hippo-Merlin Signaling Dysregulation WP45:
- PI3K-Akt Signaling Pathway WP4172
- miRNA targets in ECM and membrane recep

WikiPathways 2019 Mouse 

- PluriNetWork WP1763
- Focal Adhesion WP85
- Focal Adhesion-PI3K-Akt-mTOR-signaling pa
- Factors and pathways affecting insulin-like g
- Primary Focal Segmental Glomerulosclerosis:

KEGG 2019 Human 

- Proteoglycans in cancer
- ECM-receptor interaction
- Hematopoietic cell lineage
- Focal adhesion
- PI3K-Akt signaling pathway

ARCHS4 Kinases Coexp 

- MYLK_human_kinase_ARCHS4_coexpression
- NEK7_human_kinase_ARCHS4_coexpression
- DDR2_human_kinase_ARCHS4_coexpression
- ABL2_human_kinase_ARCHS4_coexpression
- AXL_human_kinase_ARCHS4_coexpression

KEGG 2019 Mouse 

- Proteoglycans in cancer
- ECM-receptor interaction
- Hematopoietic cell lineage
- Focal adhesion
- PI3K-Akt signaling pathway

BioCarta 2016 

- TGF beta signaling pathway_Homo sapiens_
- Insulin Signaling Pathway_Homo sapiens_h_
- Multiple antiapoptotic pathways from IGF-1
- The IGF-1 Receptor and Longevity_Homo sai
- Bone Remodeling_Homo sapiens_h_rankPat

Reactome 2016 

- Elastic fibre formation_Homo sapiens_R-HSA
- Extracellular matrix organization_Homo sap
- Crosslinking of collagen fibrils_Homo sapien
- Collagen formation_Homo sapiens_R-HSA-1+
- Assembly of collagen fibrils and other multi

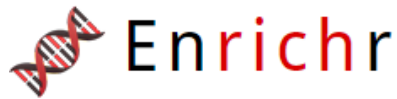
HumanCyc 2016 

- allopregnanolone biosynthesis_Homo sapie
- pyrimidine deoxyribonucleosides salvage_Hi
- urea cycle_Homo sapiens_PWWY-4984
- androgen biosynthesis_Homo sapiens_PWWY
- nicotine degradation III_Homo sapiens_PWWY

NCI-Nature 2016 

- Beta3 integrin cell surface interactions_Hom
- Beta5 beta6 beta7 and beta8 integrin cell su
- Validated transcriptional targets of TAp63 is
- Beta1 integrin cell surface interactions_Hom
- Integrins in angiogenesis_Homo sapiens_2d

Understanding the results



[Login](#) | [Register](#)

[Transcription](#) **[Pathways](#)** [Ontologies](#) [Diseases/Drugs](#) [Cell Types](#) [Misc](#) [Legacy](#) [Crowd](#)

Description No description available (171 genes)



WikiPathways 2019 Human

WikiPathways 2019 Mouse

KEGG 2019 Human

Bar Graph

Table

Clustergram



Click the bars to sort. Now sorted by p-value ranking.

[SVG](#) [PNG](#) [JPG](#)

Proteoglycans in cancer

ECM-receptor interaction

Hematopoietic cell lineage

Focal adhesion

PI3K-Akt signaling pathway

Protein digestion and absorption

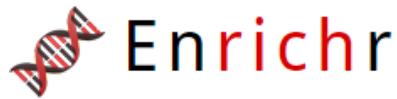
FoxO signaling pathway

Pathways in cancer

Malaria

Rap1 signaling pathway

Understanding the results



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Description No description available (171 genes)

WikiPathways 2019 Human

WikiPathways 2019 Mouse

KEGG 2019 Human

Bar Graph

[Table](#)

[Clustergram](#)



Click the bars to sort. Now sorted by p-value ranking.

[SVG](#) [PNG](#) [JPG](#)

Proteoglycans in cancer

p-value: 0.00007320
q-value: 0.01127
odds ratio: 8.56
combined score: 81.49

ECM-receptor interaction

Hematopoietic cell lineage

Epithelial cell adhesion

Pathway

and absorption

Pathway

Pathways in cancer

Malaria

Rap1 signaling pathway

Pointing each row, the enrichment scores of each term are displayed

Understanding the results



Enrichr

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Description No description available (171 genes)



WikiPathways 2019 Human

WikiPathways 2019 Mouse

KEGG 2019 Human

Bar Graph

Table

Clustergram



Hover each row to see the overlapping genes.

10 entries per page

Search:

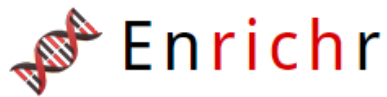
Index	Name	P-value	Adjusted p-value	Odds Ratio	Combined score
1	Proteoglycans in cancer	0.000001306	0.0004023	6.40	86.72
2	ECM-receptor interaction	0.00007320	0.01127	8.56	81.49
3	Caffeine metabolism	0.04203	0.5884	23.39	74.14
4	Hematopoietic cell lineage	0.0001859	0.01909	7.23	62.15
5	Vitamin B6 metabolism	0.05022	0.6187	19.49	58.31
6	Protein digestion and absorption	0.001054	0.05411	6.50	44.54
7	Focal adhesion	0.0003216	0.02477	4.70	37.81
8	Malaria	0.008483	0.2903	7.16	34.15
9	Ubiquinone and other terpenoid-quinone biosynthesis	0.09015	0.8167	10.63	25.58
10	FoxO signaling pathway	0.005572	0.2452	4.43	22.99

Showing 1 to 10 of 169 entries | [Export entries to table](#)

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Terms marked with an * have an overlap of less than 5

Disease/Drugs



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Transcription Pathways Ontologies **Diseases/Drugs** Cell Types Misc Legacy Crowd

Description No description available (171 genes)



PheWeb 2019



Cyst of kidney, acquired
Intestinal disaccharidase def
Endometriosis
Primary thrombocytopenia
Vitamin deficiency

ClinVar 2019



thoracic aortic aneurysm and aortic dissecti

DepMap WG CRISPR Screens Sanger CellLines



NCI-H650

DepMap WG CRISPR Screens Broad Cell

NCI-H520

RT-4

SNU-840

T3M-4

Hs 683

To test if the list of input genes is enriched in well-known genes associated to a certain disease (**disease genes**) or in **target of therapeutic drugs**

PR segment

Glaucoma

Spherical equivalent (joint analysis main effi

Distance_between_home_and_job_workplac

Left_arm_predicted_mass_23126_raw

Pulse_wave_Arterial_Stiffness_index_21021_u

DisGeNET



Neoplasm Metastasis

Tumor Progression

Endometriosis

Osteosarcoma of bone

Osteosarcoma

DSigDB



AFLATOXIN B1_CTD_00007128

estradiol_CTD_00005920

Tetradioxin_CTD_00006848

quercetin_CTD_00006679

trichostatin A_CTD_00000660

ARCHS4 IDG Coexp



NEK7_IDG_kinase_ARCHS4_coexpression

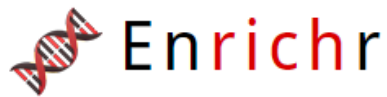
GPRC5A_IDG_GPCR_ARCHS4_coexpression

CDK15_IDG_kinase_ARCHS4_coexpression

FAM26E_IDG_ionchannel_ARCHS4_coexpres

ALPK2_IDG_kinase_ARCHS4_coexpression

Disease/Drugs



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[Transcription](#) [Pathways](#) [Ontologies](#) **[Diseases/Drugs](#)** [Cell Types](#) [Misc](#) [Legacy](#) [Crowd](#)

Description No description available (171 genes)



PheWeb 2019



Cyst of kidney, acquired
Intestinal disaccharidase deficiencies and di
Endometriosis
Primary thrombocytopenia
Vitamin deficiency

ClinVar 2019



thoracic aortic aneurysm and aortic dissecti
neural tube defect
disorder of the urea cycle metabolism
craniosynostosis syndrome
hepatocellular carcinoma

DepMap WG CRISPR Screens Sanger CellLines



NCI-H650
SW620
BxPC-3
HCC38
MCAS

DepMap WG CRISPR Screens Broad CellLines



NCI-H520
RT-4
SNU-840
T3M-4
Hs 683

GWAS Catalog 2019



Intraocular pressure
Central corneal thickness
PR segment
Glaucoma

UK Biobank GWAS v1



Impedance_of_right_leg_23107_raw
Standing_height_50_raw
Distance_between_home_and_job_workplac
Left_arm_predicted_mass_23126_raw
Pulse_wave_Arterial_Stiffness_index_21021_u

DisGeNET



Neoplasm Metastasis
Tumor Progression
Endometriosis
Osteosarcoma of bone
Osteosarcoma

To test if the list of
input genes is
enriched in
disease genes
collected in
DisGeNet

ARCHS4 IDG Coexp



NEK7_IDG_kinase_ARCHS4_coexpression
GPRC5A_IDG_GPCR_ARCHS4_coexpression
CDK15_IDG_kinase_ARCHS4_coexpression
FAM26E_IDG_ionchannel_ARCHS4_coexpres
ALPK2_IDG_kinase_ARCHS4_coexpression

[Transcription](#)
[Pathways](#)
[Ontologies](#)
[Diseases/Drugs](#)
[Cell Types](#)
[Misc](#)
[Legacy](#)
[Crowd](#)

Description No description available (171 genes)



PheWeb 2019

ClinVar 2019

DepMap WG CRISPR Screens Sanger CellLines 2019

DepMap WG CRISPR Screens Broad CellLines 2019

GWAS Catalog 2019

UK Biobank GWAS v1

DisGeNET

Bar Graph

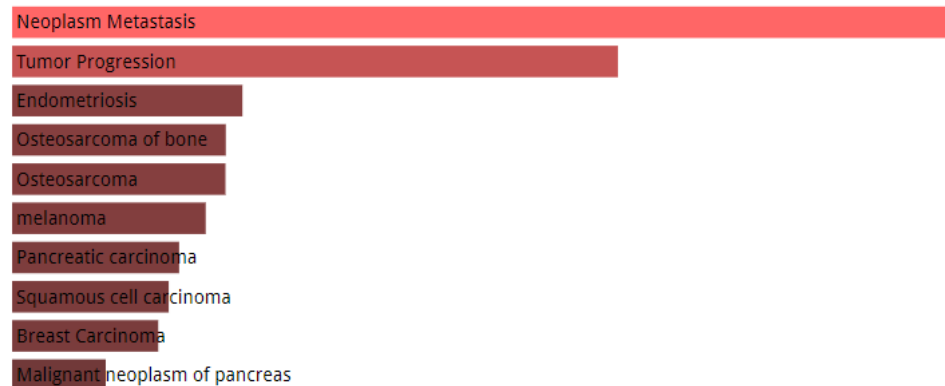
Table

Clustergram

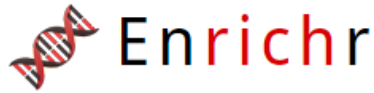


Click the bars to sort. Now sorted by **p-value ranking**.

[SVG](#) [PNG](#) [JPG](#)



Transcription



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Transcription

[Pathways](#)

[Ontologies](#)

[Diseases/Drugs](#)

[Cell Types](#)

[Misc](#)

[Legacy](#)

[Crowd](#)

Description No description available (171 genes)



ChEA 2016

CJUN_26792858_Chip-Seq_BT
SUZ12_18974828_Chip-Seq_M
SMAD2_18955904_Chip-ChIP
SMAD3_18955904_Chip-ChIP
ZNF217_24962896_Chip-Seq

ENCODE and ChEA

ARCHS4 TFs Coexp

TF Perturbations Followed by Expression

NFKB1_ACTIVATION_HUMAN_GSE20736_CRE
ESR1_KD_HUMAN_GSE27473_CREEDSID_GE
FLI1_KD_HUMAN_GSE27524_CREEDSID_GEN
PCGF2_KD_HUMAN_GSE7578_CREEDSID_GE
TP63_KD_HUMAN_GSE20286_CREEDSID_GE

TP53_human
TRP53_mouse
HIF1A_human
SP1_human
NFKB1_mouse

FOXL1
ARNTL2
PLEK2
LARP6
RUNX2

TRANSFAC and JASPAR PWMs

POU2F2 (human)
REL (human)
FOXO1 (human)
TCFAP2A (human)
STAT3 (human)

Epigenomics Roadmap HM ChIP-seq

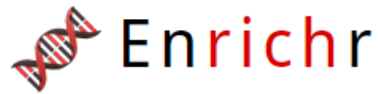
H3K27me3 H1
H3K27me3 Colonic Mucosa
H2BK20ac IMR90
H3K27me3 Fetal Brain
H2BK12ac IMR90

TargetScan microRNA 2017

mmu-miR-707
hsa-miR-1244
mmu-miR-10b
mmu-miR-10a
hsa-miR-508-3p

To test if the list of input genes is enriched in **target** of **Transcription Factors** (TF) based on the TF-target interactions stored in various databases

Transcription



[Login](#) | [Register](#)

Transcription Pathways Ontologies Diseases/Drugs Cell Types Misc Legacy Crowd

Description No description available (171 genes)



ChEA 2016



CJUN_26792858_Chip-Seq_BT549_Human
SUZ12_18974828_Chip-Seq_MESCs_Mouse
SMAD2_18955904_Chip-ChIP_HaCaT_Human
SMAD3_18955904_Chip-ChIP_HaCaT_Human
ZNF217_24962896_Chip-Seq_MCF-7_Human

ENCODE and ChEA Consensus TFs from



SUZ12_CHEA
TP53_CHEA
SMAD4_CHEA
NFE2L2_CHEA
GATA2_CHEA

ARCHS4 TFs Coexp



MXI_human_tf_ARCHS4_coexpression
FOXC2_human_tf_ARCHS4_coexpression
SNAI2_human_tf_ARCHS4_coexpression
HIF1A_human_tf_ARCHS4_coexpression
FOXL1_human_tf_ARCHS4_coexpression

TF Perturbations Followed by Expression



NFKB1_ACTIVATION_HUMAN_GSE20736_CRE
ESR1_KD_HUMAN_GSE27473_CREEDSID_GE
FLI1_KD_HUMAN_GSE27524_CREEDSID_GEN
PCGF2_KD_HUMAN_GSE7578_CREEDSID_GE
TP53_KD_HUMAN_GSE20286_CREEDSID_GE

TRRUST Transcription Factors 2019



TP53_human
TRP53_mouse
HIF1A_human
SP1_human
NFKB1_human

Enrichr Submissions TF-Gene Cooccurrence



FOXL1
ARNTL2
PLEK2
LARP6
GADD45

TRANSFAC and JASPAR PWMs



POU2F2 (human)
REL (human)
FOXO1 (human)
TCFAP2A (human)
STAT3 (human)

Epi

H3K
H3K
H2BK20ac IMR90
H3K27me3 Fetal Brain
H2BK12ac IMR90

mmu-miR-10b
mmu-miR-10a
hsa-miR-508-3p

To test if the input genes list is enriched in TF targets according to the **TRANSFAC** and **JASPAR** databases

Transcription

TRANSFAC and JASPAR PWMs

Bar Graph

Table

Grid

Network

Clustergram



Hover each row to see the overlapping g

10 entries per page

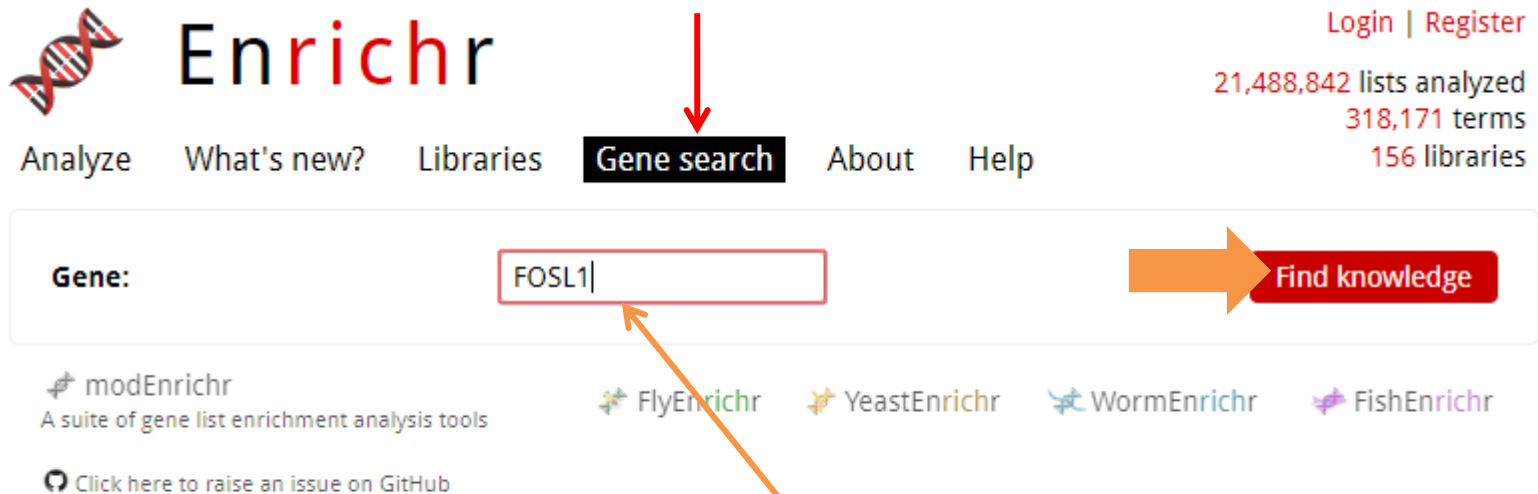
Index	Name					ined score
1	MYC::MAX (human)					20.38
2	REL (human)					11.91
3	FOXD1 (human)					11.26
4	FOXM1 (human)	0.08808	1.000	4.03		9.80
5	POU2F2 (human)	0.003967	1.000	1.55		8.57
6	Klf4 (mouse)	0.05150	1.000	2.87		8.51
7	MIR210 (human)	0.1206	1.000	3.34		7.07
8	SMAD3 (mouse)	0.1263	1.000	3.25		6.72
9	FOXO3 (human)	0.1040	1.000	2.66		6.02
10	TCFAP2A (human)	0.02485	1.000	1.63		6.01

Pointing on the TF name, you can see the names of your input genes list whose promoter regions have a consensus sequence for the TF binding

Showing 1 to 10 of 290 entries | [Export entries to table](#)

[Previous](#) [Next](#)

Gene search



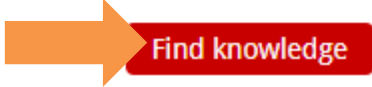
The screenshot shows the Enrichr website. At the top left is the Enrichr logo, which includes a DNA double helix icon. To the right of the logo is the word "Enrichr" in a large, bold, black font. Below the logo and name is a navigation bar with links: "Analyze", "What's new?", "Libraries", "Gene search", "About", and "Help". The "Gene search" link is highlighted with a black background and white text, and a red arrow points down to it from above. To the right of the navigation bar, there are statistics: "21,488,842 lists analyzed", "318,171 terms", and "156 libraries". Above these statistics are links for "Login" and "Register". Below the navigation bar is a search bar with the label "Gene:" and a text input field containing "FOSL1". To the right of the input field is a red button with a white arrow pointing right and the text "Find knowledge". Below the search bar, there are four icons and labels for different enrichment tools: "modEnrichr" (a green leaf icon), "FlyEnrichr" (a green fly icon), "YeastEnrichr" (a yellow star icon), "WormEnrichr" (a blue worm icon), and "FishEnrichr" (a purple fish icon). Below these icons is a link that says "Click here to raise an issue on GitHub". An orange arrow points from a text box at the bottom to the "FOSL1" text in the search input field.

Enrichr

Login | Register

21,488,842 lists analyzed
318,171 terms
156 libraries

Analyze What's new? Libraries **Gene search** About Help

Gene: 

 modEnrichr
A suite of gene list enrichment analysis tools

 FlyEnrichr  YeastEnrichr  WormEnrichr  FishEnrichr

 Click here to raise an issue on GitHub

You can search for a single gene and the categories in which it is involved (for which is annotated)



Gene:

FOSL1

[Find knowledge](#)**⊕ Transcription****⊖ Pathways****⊕ BioPlanet 2019****⊕ WikiPathways 2019 Human****⊕ WikiPathways 2019 Mouse****⊖ KEGG 2019 Human**

FOSL1 is a member of the **Human T-cell leukemia virus 1 infection** pathway.

FOSL1 is a member of the **IL-17 signaling pathway** pathway.

FOSL1 is a member of the **Osteoclast differentiation** pathway.

FOSL1 is a member of the **Wnt signaling pathway** pathway.

⊕ ARCHS4 Kinases Coexp**⊕ KEGG 2019 Mouse****⊕ BioCarta 2016****⊕ NCI-Nature 2016****⊕ Panther 2016****⊕ PPI Hub Proteins****⊕ LINCS L1000 Kinase Perturbations down****⊕ LINCS L1000 Kinase Perturbations up****⊕ Kinase Perturbations from GEO down****⊕ Kinase Perturbations from GEO up****⊕ NURSA Human Endogenous Complexome****⊕ SubCell BarCode****⊕ Ontologies****⊕ Diseases/Drugs****⊕ Cell Types****⊕ Misc****⊕ Legacy****⊕ Crowd**

enrichR knowleged graph



Enrichr

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67,462,338 sets analyze

494,081 term

225 librarie

Analyze

[What's new?](#)

[Libraries](#)

[Gene search](#)

[Term search](#)

[About](#)

[Help](#)

Input data

Expand a gene, a term, or a variant into a gene set:

e.g. STAT3, breast cancer, or rs28897756



Try an example

Include the top 100 most relevant genes



Paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background**.

Paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

0 gene(s) entered

In order to enable others to search your set please enter a brief description of it.

To create a graph that integrate different databases (e.g., KEGG, GO) linked to the genes enriched in each one

Submit

try Enrichr-KG

Try

Rummagene

<https://maayanlab.cloud/enrichr-kg>



Project Achilles



Submit your gene set for enrichment analysis with **Enrichr**

Paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

Insert your gene list

Submit

[Try an example](#)

Description

Minimum libraries per gene 1

Minimum links per gene 1

Minimum links per term 1

Select maximum of five libraries
(Scroll for more)

- ☒ GO Biological Process 2021
- ☐ CCLE Proteomics 2020
- ☐ Human Phenotype Ontology
- ☐ GWAS Catalog 2019
- ☐ PFOCR Pathways
- ☒ MGI Mammalian Phenotype Level 4 2021
- ☐ Proteomics Drug Atlas 2023
- ☐ Reactome 2022
- ☐ The Kinase Library 2023

Top terms to include

5

5

Search an Enrichr term and expand it to a gene set:

Select the databases
and the number of
top-enriched terms to
consider



Input Gene Set

LEGEND



Sample Input



Each node belonging to a category is colored accordingly (e.g., input genes are green, GO BP are pink, ..)